

Raw Data Analysis Review

Abigael Jepyego

2026-07-10

```
getwd()
```

```
## [1] "C:/Users/Lenovo/OneDrive/Documents"
```

```
list.files()
```

```
## [1] "Airport"
## [2] "Airport Analysis Review.Rmd"
## [3] "Custom Office Templates"
## [4] "desktop.ini"
## [5] "I am sharing ID ABBYH.pdf with you from WPS Office.txt"
## [6] "KingsoftData"
## [7] "Power BI Desktop"
## [8] "Raw-Data-Analysis---Review.pdf"
## [9] "Raw-Data-Analysis---Review.Rmd"
## [10] "Raw-Data-Analysis---Review_files"
## [11] "Raw Data Analysis Review.Rmd"
## [12] "RAW DATA BY AIRPORT.xlsx"
```

```
airport <- read_excel("C:/Users/Lenovo/Downloads/RAW DATA BY AIRPORT.xlsx")
```

```
## Warning: Expecting numeric in 01286 / R1286C15: got '0'
```

```
## Warning: Expecting numeric in M1403 / R1403C13: got '--'
```

```
##View data
```

```
head(airport)
```

```
## # A tibble: 6 x 24
##   Aerodrome      FY      Year Month 'Landings INT' 'Takeoffs INT' 'Total INT'
##   <chr>          <chr>    <dbl> <chr>          <dbl> <chr>          <dbl>
## 1 Eldoret Airstrip 2016/2~ 2017 Augu~          29 16          45
## 2 Eldoret Airstrip 2019/2~ 2020 Janu~           0 0           0
## 3 Eldoret Airstrip 2019/2~ 2020 Febr~           0 0           0
## 4 Eldoret Airstrip 2019/2~ 2020 March           0 0           0
## 5 Eldoret Airstrip 2019/2~ 2020 April           0 0           0
## 6 Eldoret Airstrip 2019/2~ 2020 May            0 0           0
## # i 17 more variables: 'Landings DOM' <dbl>, 'Takeoffs DOM' <dbl>,
```

```
## # 'Total DOM' <dbl>, 'TOTAL AIRCRAFT' <dbl>, 'Arrival INT' <dbl>,
## # 'Departure INT' <dbl>, 'Transit INT' <dbl>, 'TOTAL INT' <dbl>,
## # 'Arrival DOM' <dbl>, 'Departure DOM' <dbl>, 'Transit DOM' <chr>,
## # 'TOTAL DOM2' <dbl>, 'TOTAL PAX' <dbl>, TRANSIT <dbl>, Imports <chr>,
## # Exports <chr>, 'TOTAL CARGO' <chr>
```

```
str(airport)
```

```
## tibble [1,588 x 24] (S3: tbl_df/tbl/data.frame)
## $ Aerodrome      : chr [1:1588] "Eldoret Airstrip" "Eldoret Airstrip" "Eldoret Airstrip" "Eldoret Ai
## $ FY            : chr [1:1588] "2016/2017" "2019/2020" "2019/2020" "2019/2020" ...
## $ Year          : num [1:1588] 2017 2020 2020 2020 2020 ...
## $ Month         : chr [1:1588] "August" "January" "February" "March" ...
## $ Landings INT  : num [1:1588] 29 0 0 0 0 0 0 0 27 0 ...
## $ Takeoffs INT  : chr [1:1588] "16" "0" "0" "0" ...
## $ Total INT     : num [1:1588] 45 0 0 0 0 0 0 0 34 0 ...
## $ Landings DOM  : num [1:1588] 297 25 13 7 19 28 18 20 189 16 ...
## $ Takeoffs DOM  : num [1:1588] 313 23 13 7 18 27 18 21 210 15 ...
## $ Total DOM     : num [1:1588] 610 48 26 14 37 55 36 41 399 31 ...
## $ TOTAL AIRCRAFT: num [1:1588] 655 48 26 14 37 55 36 41 433 31 ...
## $ Arrival INT   : num [1:1588] 6 0 0 0 0 0 0 0 0 0 ...
## $ Departure INT : num [1:1588] 6 0 0 0 0 0 0 0 0 0 ...
## $ Transit INT   : num [1:1588] 3 0 0 0 0 0 0 0 0 0 ...
## $ TOTAL INT     : num [1:1588] 15 0 0 0 0 0 0 0 0 0 ...
## $ Arrival DOM   : num [1:1588] 7225 23 27 8 27 ...
## $ Departure DOM : num [1:1588] 6894 38 10 10 18 ...
## $ Transit DOM   : chr [1:1588] "2860" "7" "7" "0" ...
## $ TOTAL DOM2    : num [1:1588] 16979 68 44 18 45 ...
## $ TOTAL PAX     : num [1:1588] 16994 68 44 18 45 ...
## $ TRANSIT       : num [1:1588] 2863 7 7 0 0 ...
## $ Imports       : chr [1:1588] "966504" "0" "0" "0" ...
## $ Exports       : chr [1:1588] "-" "0" "0" "0" ...
## $ TOTAL CARGO   : chr [1:1588] "966504" "0" "0" "0" ...
```

```
summary(airport)
```

```
##      Aerodrome      FY      Year      Month
## Length :1588 Length :1588 Min. :2017 Length :1588
## N.unique : 22 N.unique : 10 1st Qu.:2020 N.unique : 16
## N.blank : 0 N.blank : 0 Median :2022 N.blank : 0
## Min.nchar: 6 Min.nchar: 9 Mean :2022 Min.nchar: 3
## Max.nchar: 26 Max.nchar: 9 3rd Qu.:2024 Max.nchar: 9
##      NAs      : 23 Max. :2026
##      NAs      :23
## Landings INT Takeoffs INT Total INT Landings DOM
## Min. : 0.0 Length :1588 Min. : 0.0 Min. : 0.0
## 1st Qu.: 0.0 N.unique : 287 1st Qu.: 0.0 1st Qu.: 33.0
## Median : 0.0 N.blank : 0 Median : 0.0 Median : 211.5
## Mean : 206.7 Min.nchar: 1 Mean : 413.8 Mean : 537.0
## 3rd Qu.: 53.0 Max.nchar: 4 3rd Qu.: 94.0 3rd Qu.: 503.0
## Max. :3490.0 NAs : 156 Max. :7370.0 Max. :24301.0
## NAs :155 NAs :152 NAs :158
## Takeoffs DOM Total DOM TOTAL AIRCRAFT Arrival INT
```

```
## Min. : 0.0 Min. : 0.0 Min. : 0 Min. : 0.00
## 1st Qu.: 33.0 1st Qu.: 65.5 1st Qu.: 72 1st Qu.: 0.00
## Median : 253.0 Median : 460.0 Median : 493 Median : 0.00
## Mean : 518.8 Mean : 1052.6 Mean : 1402 Mean : 14652.88
## 3rd Qu.: 512.5 3rd Qu.: 1007.2 3rd Qu.: 1024 3rd Qu.: 82.25
## Max. :4942.0 Max. :24822.0 Max. :24828 Max. :337059.00
## NAs :157 NAs :152 NAs :79 NAs :156
## Departure INT Transit INT TOTAL INT Arrival DOM
## Min. : 0 Min. : 0 Min. : 0 Min. : 0
## 1st Qu.: 0 1st Qu.: 0 1st Qu.: 0 1st Qu.: 101
## Median : 0 Median : 0 Median : 1 Median : 2603
## Mean : 13387 Mean : 2315 Mean : 30268 Mean : 12287
## 3rd Qu.: 69 3rd Qu.: 2 3rd Qu.: 398 3rd Qu.: 12416
## Max. :347873 Max. :127912 Max. :705965 Max. :102006
## NAs :159 NAs :175 NAs :155 NAs :157
## Departure DOM Transit DOM TOTAL DOM2 TOTAL PAX
## Min. : 0.0 Length :1588 Min. : 0.0 Min. : 0
## 1st Qu.: 102.5 N.unique : 638 1st Qu.: 193.5 1st Qu.: 399
## Median : 2866.0 N.blank : 0 Median : 5771.0 Median : 6813
## Mean : 12088.5 Min.nchar: 1 Mean : 24789.3 Mean : 58948
## 3rd Qu.: 12094.5 Max.nchar: 5 3rd Qu.: 25936.0 3rd Qu.: 27352
## Max. :107704.0 NAs : 201 Max. :204911.0 Max. :905283
## NAs :157 NAs :145 NAs :71
## TRANSIT Imports Exports TOTAL CARGO
## Min. : 0 Length :1588 Length :1588 Length :1588
## 1st Qu.: 0 N.unique : 266 N.unique : 227 N.unique : 322
## Median : 154 N.blank : 0 N.blank : 0 N.blank : 0
## Mean : 3950 Min.nchar: 1 Min.nchar: 1 Min.nchar: 1
## 3rd Qu.: 1268 Max.nchar: 18 Max.nchar: 18 Max.nchar: 18
## Max. :147828 NAs : 341 NAs : 341 NAs : 306
## NAs :131
```

#Check for missings values

```
colSums(is.na(airport))
```

```
## Aerodrome FY Year Month Landings INT
## 0 23 23 0 155
## Takeoffs INT Total INT Landings DOM Takeoffs DOM Total DOM
## 156 152 158 157 152
## TOTAL AIRCRAFT Arrival INT Departure INT Transit INT TOTAL INT
## 79 156 159 175 155
## Arrival DOM Departure DOM Transit DOM TOTAL DOM2 TOTAL PAX
## 157 157 201 145 71
## TRANSIT Imports Exports TOTAL CARGO
## 131 341 341 306
```

#Create a proper column data

```
airport$Date <- as.Date(
  paste(
    airport$Year,
    airport$Month,
```

```

    "1"
  ),
  format="%Y %B %d"
)

```

```
#Sort the data
```

```

library(dplyr)
airport <- airport %>%
  arrange(Date)

```

```
names(airport)
```

```

## [1] "Aerodrome"      "FY"              "Year"            "Month"
## [5] "Landings INT"   "Takeoffs INT"    "Total INT"        "Landings DOM"
## [9] "Takeoffs DOM"   "Total DOM"        "TOTAL AIRCRAFT"   "Arrival INT"
## [13] "Departure INT"  "Transit INT"      "TOTAL INT"        "Arrival DOM"
## [17] "Departure DOM"  "Transit DOM"      "TOTAL DOM2"       "TOTAL PAX"
## [21] "TRANSIT"        "Imports"          "Exports"          "TOTAL CARGO"
## [25] "Date"

```

```
colnames(airport)
```

```

## [1] "Aerodrome"      "FY"              "Year"            "Month"
## [5] "Landings INT"   "Takeoffs INT"    "Total INT"        "Landings DOM"
## [9] "Takeoffs DOM"   "Total DOM"        "TOTAL AIRCRAFT"   "Arrival INT"
## [13] "Departure INT"  "Transit INT"      "TOTAL INT"        "Arrival DOM"
## [17] "Departure DOM"  "Transit DOM"      "TOTAL DOM2"       "TOTAL PAX"
## [21] "TRANSIT"        "Imports"          "Exports"          "TOTAL CARGO"
## [25] "Date"

```

```
head(airport)
```

```

## # A tibble: 6 x 25
##   Aerodrome      FY      Year Month 'Landings INT' 'Takeoffs INT' 'Total INT'
##   <chr>          <chr> <dbl> <chr>          <dbl> <chr>          <dbl>
## 1 Eldoret Int. Airp~ 2016~ 2017 Janu~          18 10          28
## 2 Jomo Kenyatta Int~ 2016~ 2017 Janu~        2909 3275        6184
## 3 Kisumu Int. Airpo~ 2016~ 2017 Janu~           5 5          10
## 4 Lokichoggio Airpo~ 2016~ 2017 Janu~          41 40          81
## 5 Malindi Airport    2016~ 2017 Janu~           5 3           8
## 6 Manda Airstrip      2016~ 2017 Janu~           0 0           0
## # i 18 more variables: 'Landings DOM' <dbl>, 'Takeoffs DOM' <dbl>,
## #   'Total DOM' <dbl>, 'TOTAL AIRCRAFT' <dbl>, 'Arrival INT' <dbl>,
## #   'Departure INT' <dbl>, 'Transit INT' <dbl>, 'TOTAL INT' <dbl>,
## #   'Arrival DOM' <dbl>, 'Departure DOM' <dbl>, 'Transit DOM' <chr>,
## #   'TOTAL DOM2' <dbl>, 'TOTAL PAX' <dbl>, TRANSIT <dbl>, Imports <chr>,
## #   Exports <chr>, 'TOTAL CARGO' <chr>, Date <date>

```

```
#Check for missing years
```

```
sum(is.na(airport$Year))
```

```
## [1] 23
```

```
#Remove rows with missing values
```

```
airport <- airport[!is.na(airport$Year), ]  
airport <- airport[!is.na(airport$`TOTAL PAX`), ]
```

```
sum(is.na(airport$Year))
```

```
## [1] 0
```

```
sum(is.na(airport$`TOTAL PAX`))
```

```
## [1] 0
```

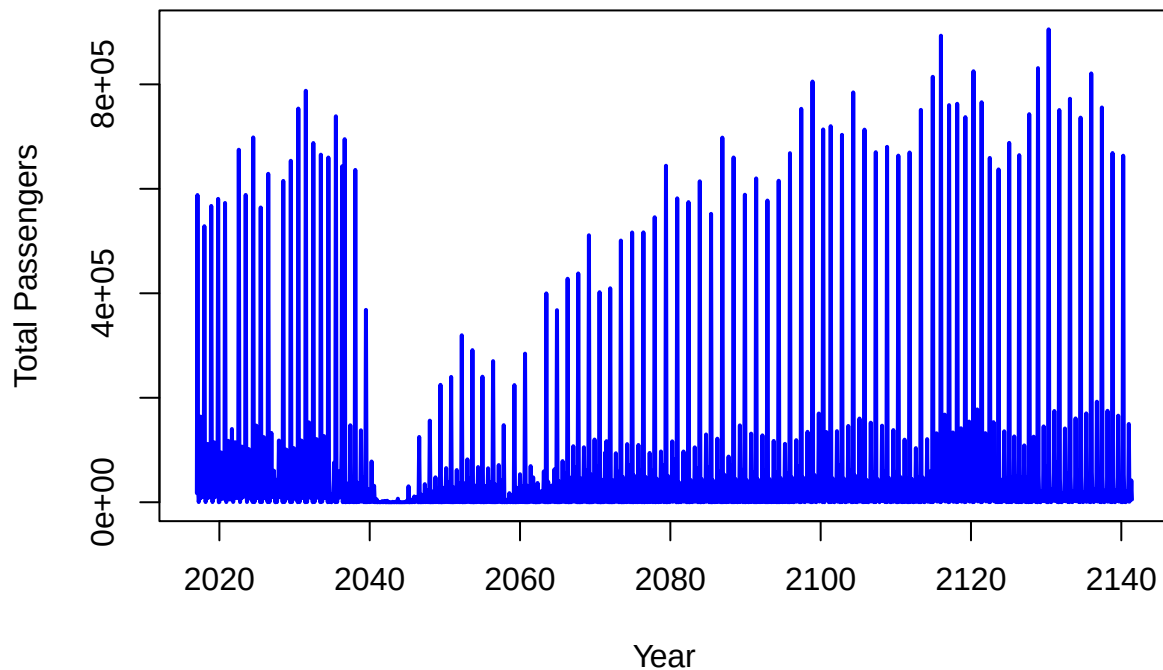
```
#Create a time series
```

```
passengers <- airport$`TOTAL PAX`  
  
ts_passengers <- ts(  
  passengers,  
  start = c(min(airport$Year), 1),  
  frequency = 12  
)
```

```
#Plot a time series
```

```
plot(  
  ts_passengers,  
  main = "Monthly Passenger Traffic",  
  xlab = "Year",  
  ylab = "Total Passengers",  
  col = "blue",  
  lwd = 2  
)
```

Monthly Passenger Traffic



```
adf.test(ts_passengers)
```

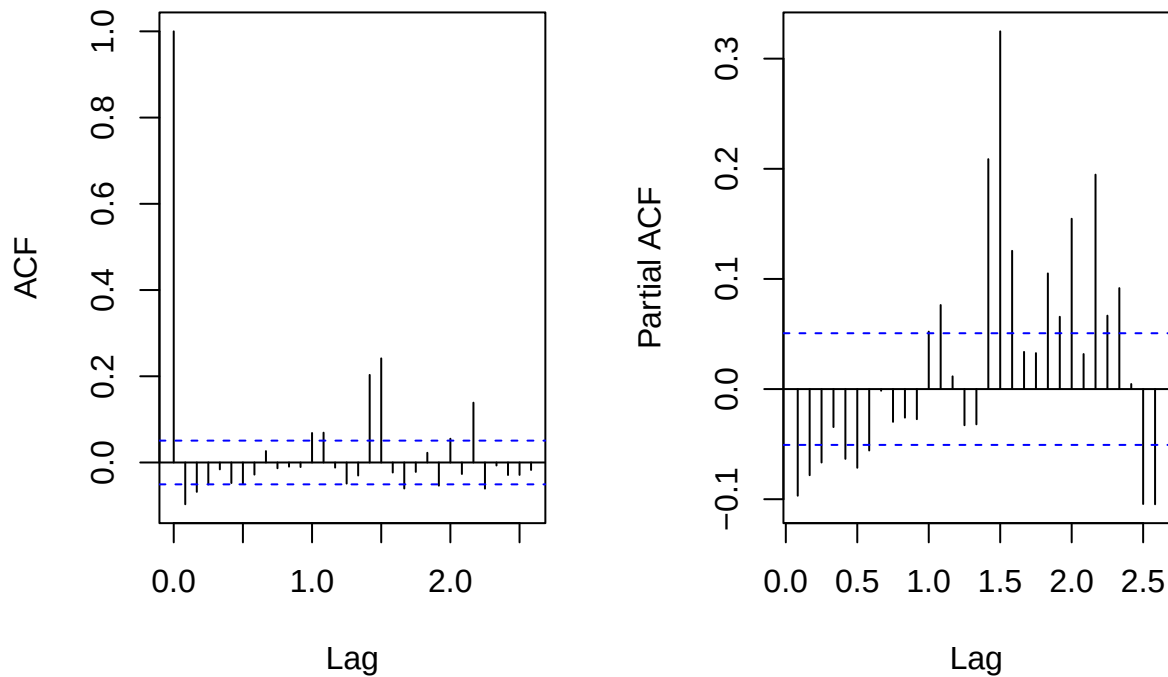
```
## Warning in adf.test(ts_passengers): p-value smaller than printed p-value
```

```
##  
## Augmented Dickey-Fuller Test  
##  
## data: ts_passengers  
## Dickey-Fuller = -12.947, Lag order = 11, p-value = 0.01  
## alternative hypothesis: stationary
```

```
#Plot the ACF and PACF
```

```
par(mfrow = c(1, 2))  
  
acf(ts_passengers,  
    main = "Autocorrelation Function (ACF)")  
  
pacf(ts_passengers,  
     main = "Partial Autocorrelation Function (PACF)")
```

Autocorrelation Function (ACF) Partial Autocorrelation Function (PACF)



```
par(mfrow = c(1, 1))
```

```
#Fitting the Arima Model
```

```
library(forecast)
```

```
model <- auto.arima(ts_passengers)
```

```
summary(model)
```

```
## Series: ts_passengers
## ARIMA(5,1,2)(2,0,1)[12]
##
## Coefficients:
##          ar1      ar2      ar3      ar4      ar5      ma1      ma2      sar1
##          0.6789 -0.0067 -0.0413 -0.0044 -0.1025 -1.9058  0.9186 -0.5912
## s.e.      0.0290  0.0315  0.0316  0.0314  0.0272  0.0135  0.0140  0.0688
##          sar2      sma1
##          -0.0813  0.6625
## s.e.      0.0307  0.0650
##
## sigma^2 = 1.887e+10: log likelihood = -19779.3
## AIC=39580.6   AICc=39580.77   BIC=39638.99
##
## Training set error measures:
```

```
##           ME      RMSE      MAE MPE MAPE      MASE      ACF1
## Training set -1035.742 136848.9 78460.03 NaN  Inf 0.9176227 -0.003616769
```

```
#Forecast for the next 12 months
```

```
forecast_12 <- forecast(model, h = 12)
```

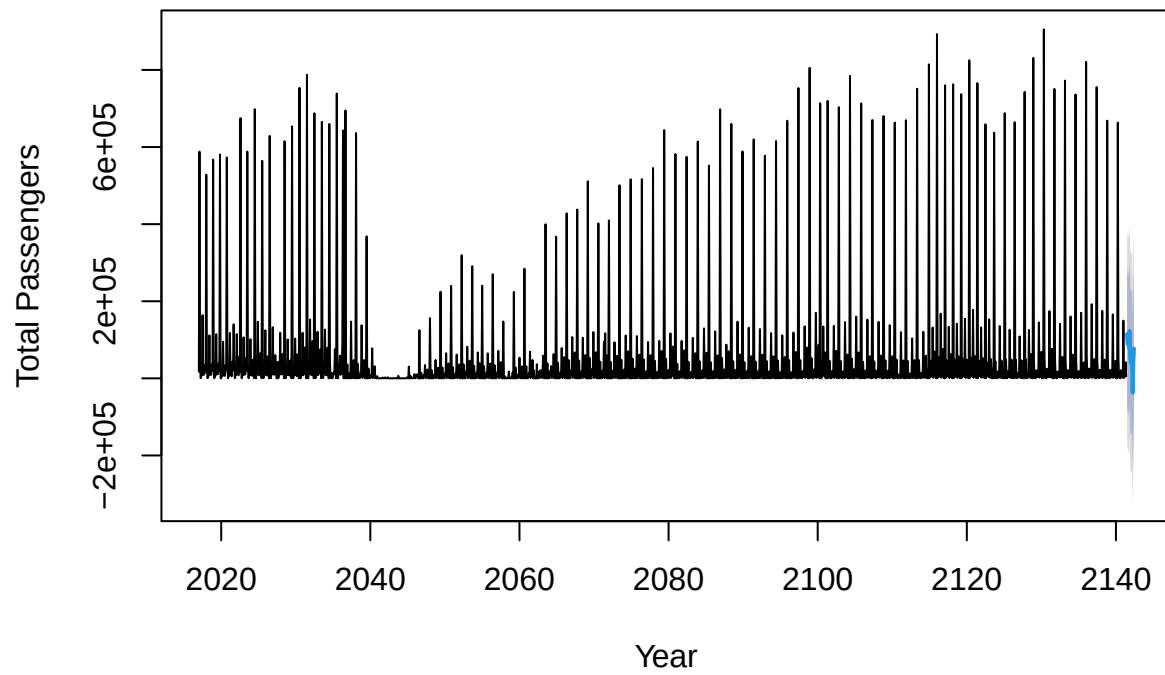
```
forecast_12
```

```
##           Point Forecast      Lo 80      Hi 80      Lo 95      Hi 95
## Jul 2141      114678.48 -61349.67 290706.6 -154533.3 383890.3
## Aug 2141      88937.71 -91565.21 269440.6 -187117.7 364993.1
## Sep 2141      95633.48 -86737.13 278004.1 -183278.3 374545.2
## Oct 2141      79671.91 -104071.09 263414.9 -201338.8 360682.6
## Nov 2141     123045.70 -61081.99 307173.4 -158553.3 404644.7
## Dec 2141      53068.07 -132423.87 238560.0 -230617.4 336753.5
## Jan 2142      38959.82 -146694.40 224614.0 -244973.8 322893.5
## Feb 2142      42705.62 -142949.05 228360.3 -241228.7 326639.9
## Mar 2142      40779.98 -144967.66 226527.6 -243296.5 324856.5
## Apr 2142     -37005.26 -222920.49 148910.0 -321338.1 247327.5
## May 2142      44380.13 -141798.75 230559.0 -240355.9 329116.1
## Jun 2142      77371.07 -109049.44 263791.6 -207734.5 362476.6
```

```
#Plot the forecast
```

```
plot(
  forecast_12,
  main = "12-Month Passenger Forecast",
  xlab = "Year",
  ylab = "Total Passengers"
)
```

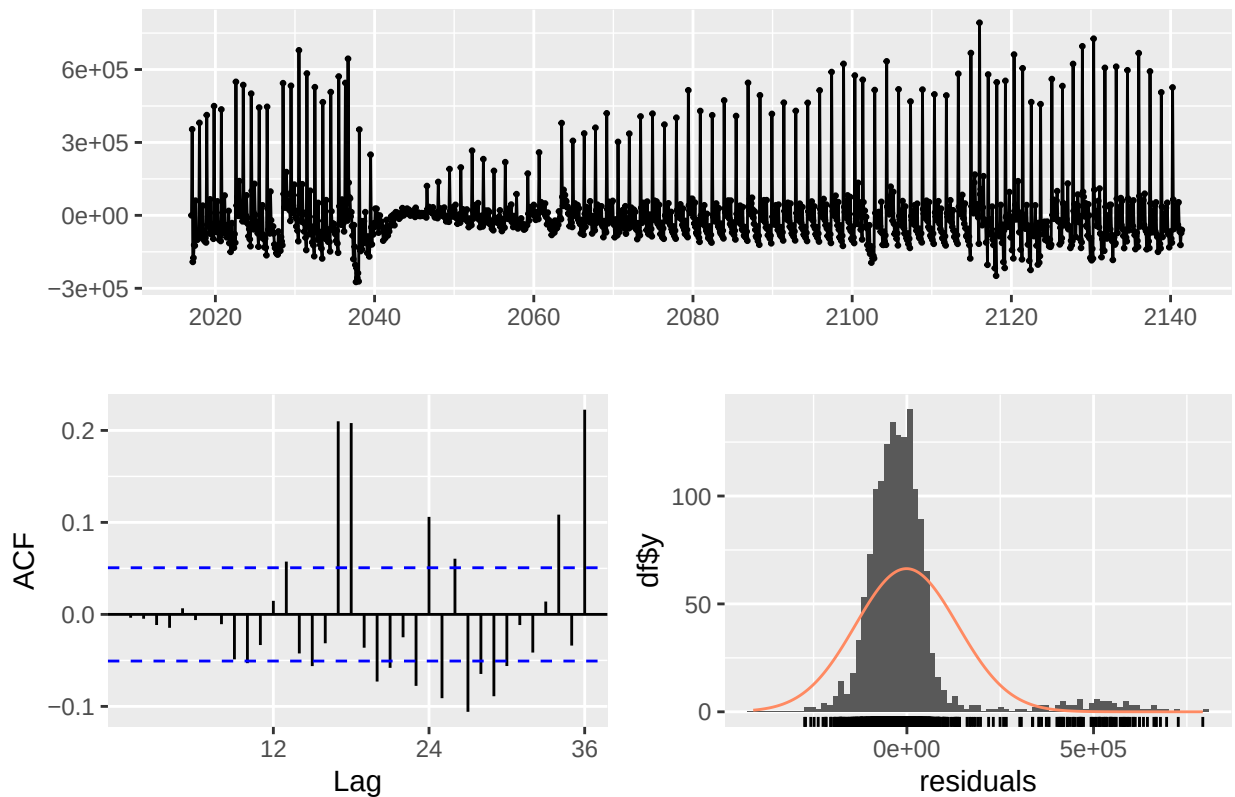

12-Month Passenger Forecast



#Evaluate the model

```
checkresiduals(model)
```

Residuals from ARIMA(5,1,2)(2,0,1)[12]



```
##
##  Ljung-Box test
##
## data:  Residuals from ARIMA(5,1,2)(2,0,1)[12]
## Q* = 199.22, df = 14, p-value < 2.2e-16
##
## Model df: 10.   Total lags used: 24
```

```
summary(model)
```

```
## Series: ts_passengers
## ARIMA(5,1,2)(2,0,1)[12]
##
## Coefficients:
##      ar1      ar2      ar3      ar4      ar5      ma1      ma2      sar1
##      0.6789 -0.0067 -0.0413 -0.0044 -0.1025 -1.9058  0.9186 -0.5912
## s.e.  0.0290  0.0315  0.0316  0.0314  0.0272  0.0135  0.0140  0.0688
##      sar2      sma1
##      -0.0813  0.6625
## s.e.  0.0307  0.0650
##
## sigma^2 = 1.887e+10:  log likelihood = -19779.3
## AIC=39580.6   AICc=39580.77   BIC=39638.99
##
## Training set error measures:
```

```
##              ME      RMSE      MAE MPE MAPE      MASE      ACF1
## Training set -1035.742 136848.9 78460.03 NaN  Inf  0.9176227 -0.003616769
```

```
#Model Summary
```

```
The best model selected by auto.arima() was:
```

```
ARIMA(5,1,5)(0,0,2)[12]**
```